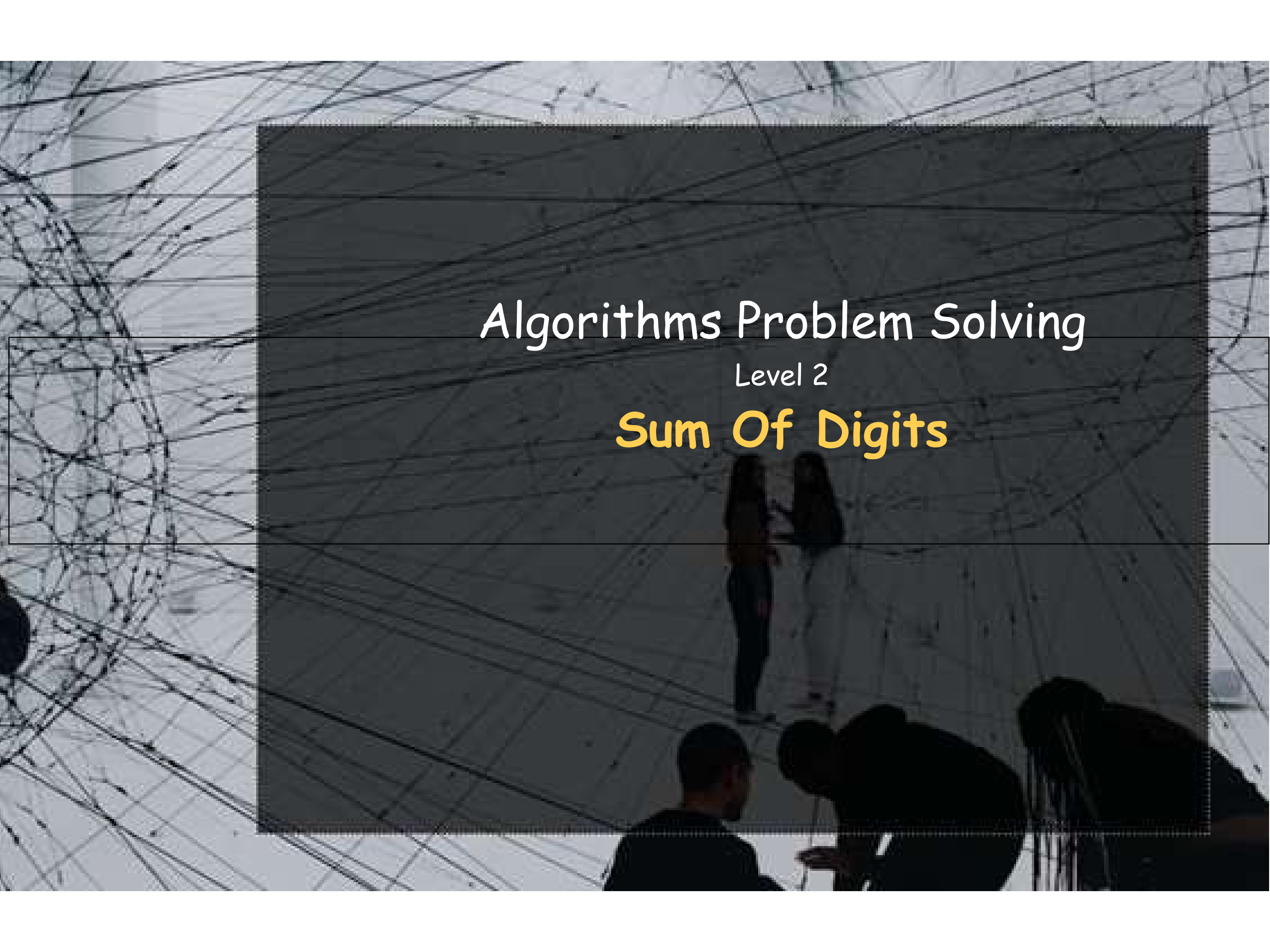


Algorithms Problem Solving

Level 2

Sum Of Digits



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Definition

- ◆ The sum of digits means adding all the digits of a number together.
- ◆ For example, ` $1234 = 1 + 2 + 3 + 4 = 10`$ '.

Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``SumOfDigits()`` extracts each digit and adds it to ``Sum``.
- ◆ ``N % 10`` gets the last digit of ``N``.
- ◆ ``N = N / 10`` removes the last digit from ``N``.
- ◆ ``Sum = Sum + Remainder`` adds the extracted digit to the total sum.
- ◆ The loop continues until ``N`` becomes ``0``.
- ◆ ``SumOfDigits()`` returns the final sum.
- ◆ ``PrintResult()`` calls ``SumOfDigits()`` and prints the result.

Steps in main()

1. Read a positive number N.
2. Get the last digit using `N % 10`.
3. Remove the last digit using `N / 10`.
4. Add the extracted digit to `Sum`.
5. Repeat until N becomes `0`.
6. Return and print the final sum.

💡 Example

◆ Input:

`1234`

◆ ` $1234 \% 10 = 4$ ` → Sum = 4

◆ ` $123 / 10$ ` after removing 4

◆ ` $123 \% 10 = 3$ ` → Sum = 7

◆ ` $12 \% 10 = 2$ ` → Sum = 9

◆ ` $1 \% 10 = 1$ ` → Sum = 10

➡ Result

`Sum Of Digits = 10`

Note

- ◆ The same technique from the previous lesson is used here: `` % 10`` extracts the last digit and `` / 10`` removes it.
- ◆ The difference is that instead of printing each digit, the program adds every digit to ``Sum``.

⊘ Warning

- ◆ `Sum` must be initialized to `0`; otherwise, it would contain an undefined value and the calculation would be incorrect.
- ◆ The program only accepts positive numbers because of the validation in `ReadNumber()`.



Comparison

▨ Print Digits In Reversed Order

- ✓ Extracts each digit and prints it.
- ✓ Example: `1234 → 4 3 2 1`

▨ Sum Of Digits

- ✓ Extracts each digit and adds it to `Sum`.
- ✓ Example: `1234 → 4 + 3 + 2 + 1 = 10`

✦ Summary

- ✓ `` % 10`` extracts the last digit.
- ✓ `` / 10`` removes the last digit.
- ✓ Each digit is added to ``Sum``.
- ✓ The loop continues until N becomes ``0``.
- ✓ ``SumOfDigits()`` returns the sum of all digits.

Thank you 😊